Tutorial 2 - Backtesting, Regime Filters, and Basket-Based Mean Reversion

How to turn a relative-value idea into a disciplined research workflow.

Session anchor: April 23, 2026 session: z-score, spread, false signals, short squeeze, basket value, paper trading only.

Audience	Students moving from discretionary chart reading into simple model-driven research.
Last updated	April 23, 2026
External depth	NIST statistics guidance, academic stat-arb literature, and TradingView official docs added to the session framework.

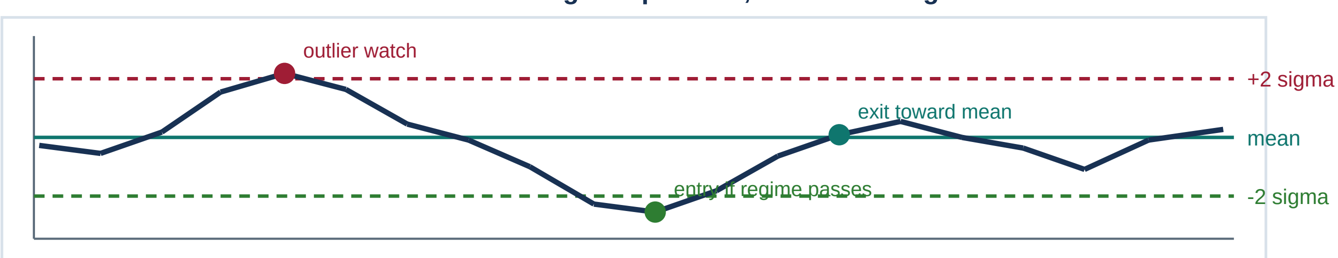
What students should be able to do after this tutorial

- Define fair basket value, spread, z-score, and regime filter in plain English.
- Build a simple mean-reversion workflow without pretending every divergence must revert.
- Add kill switches for event days, squeezes, and directional regimes.
- Separate backtest design, paper trading, and live deployment into three different stages.

Core coaching principle

The session repeatedly warned against using unfinished models with real money. Keep that rule. Backtests are laboratories, paper trading is rehearsal, and live trading is a separate decision.

Mean reversion is a regime question, not a blind signal.



1. What the session was really teaching

The conversation was not just about an algorithm. It was teaching a way to think: build a basket, estimate fair value, measure divergence, and then ask a harder question - is this still a mean-reversion regime, or has the market become directional?

That distinction matters because a model can be mathematically correct and still lose money on the wrong day. In the session, the specific failure mode was a short squeeze that created false signals. Students need to hear this early: the problem is often not the math alone. The problem is the regime.

2. The minimum vocabulary students need

Term	Plain-English meaning	Why it matters
Fair basket value	An estimated value implied by a basket of related assets or factors	Gives a relative anchor instead of guessing absolute direction
Spread	The gap between the traded asset and the model's fair value estimate	This is what you monitor, not just raw price
Z-score	How many standard deviations the spread is from its average	Turns a raw gap into a comparable signal
Mean-reversion regime	A market state where stretched spreads tend to move back toward normal	Lets contrarian ideas work
Directional regime	A market state where stretched spreads can stay stretched or extend further	Tells you not to fade every move
Kill switch	A rule that blocks entries when the environment is abnormal	Protects the model from event days and squeezes

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spread_t = traded_asset_t - model_value_t
z_t      = (spread_t - mean(spread)) / std(spread)
```

Statistical foundation

NIST defines a z-score as the number of standard deviations a data point is away from the mean. That is the right student-level definition: a standard way to measure how unusual today's spread is relative to normal behavior.

3. A professional workflow students can follow

Step 1 - Choose an economic story, not random tickers

A basket should exist for a reason. If the asset and its basket do not share an economic relationship, the model becomes numerology.

Step 2 - Build the basket

The session used baskets such as QQQ, Microsoft, Apple, Nvidia, and Amazon. In practice, the basket can be sector ETFs, highly related stocks, or a small set of factors that explain most of the asset's movement.

Step 3 - Normalize and compute the spread

Use consistent sampling intervals, clean missing data, and measure the difference between the asset and the model-implied value.

Step 4 - Translate spread into z-score

A raw spread of 1.5 means nothing by itself. A z-score tells you whether 1.5 is ordinary or extreme for this relationship.

Step 5 - Add the regime filter

This is the part most beginners skip. Ask whether the environment still favors mean reversion. If volatility explodes, news breaks, or a squeeze is underway, the model should step aside.

Step 6 - Paper trade the full process

Only after the rules are stable should the student use simulated orders. A paper trade is not a toy. It is where execution assumptions, patience, and review discipline are tested.

4. What the academic and platform sources add

Avellaneda and Lee's classic statistical-arbitrage paper gives students a solid bridge from the session to formal research. Their framework studies model-driven stat-arb in U.S. equities using PCA or sector ETFs, then models residual or idiosyncratic returns as mean-reverting processes that naturally generate contrarian signals.

TradingView's official Pine documentation explains that scripts can request data from other symbols and timeframes. That matters because students often prototype baskets by pulling related instruments into one script. TradingView's official Paper Trading documentation is equally useful: it frames paper trading as a risk-free simulator for applying technical and fundamental analysis without real-money risk.

5. The regime filter checklist

Question	If YES	If NO
Is today an event day or headline day?	Reduce size or skip new entries	Continue to next filter
Has realized volatility jumped sharply?	Widen assumptions or block mean-reversion entries	Continue to next filter
Is there evidence of a squeeze or one-way flow?	Kill switch on; review after the move settles	Continue to next filter
Are correlations breaking down across the basket?	Re-estimate or suspend the model	Continue to next filter
Did the spread move because the model is wrong, or because the market is stressed?	Do not trade until that distinction is clear	Only then consider paper execution

Rule students should memorize

A high z-score is not an order. It is a question. The real question is whether this market still behaves like a mean-reversion market.

6. Common mistakes and how to correct them

Mistake 1: Overfitting the entry threshold to the past. Correction: simplify the rule and test it across more than one regime.

Mistake 2: Ignoring transaction costs and slippage. Correction: include them before you trust the backtest.

Mistake 3: Treating every outlier as an opportunity. Correction: outliers created by news, squeezes, or structural breaks often need a kill switch, not courage.

Mistake 4: Jumping from chart idea to real money. Correction: move in sequence - backtest, paper trade, review, then decide whether live deployment is justified.

7. Student build plan

Stage	Student task	Output
Research	Pick one asset and a basket with a real economic relationship	1-page model note
Model	Compute spread and z-score on a clean sample	Chart plus parameter sheet
Filter	Write a regime checklist and at least one kill-switch rule	Checklist students can follow by hand
Paper trade	Run 20 simulated trades with fixed entry and exit rules	Trade log with screenshots and comments
Review	Separate model mistakes from execution mistakes	Revised playbook

Completion standard

A student passes this tutorial when they can explain why a model worked, why it failed, and why a paper-trading sample is still not the same thing as production deployment.

Selected external sources used in this tutorial

1. NIST - Statistical Detection of Outliers in the Certification of NIST Reference Materials (z-score definition) - https://tsapps.nist.gov/publication/get_pdf.cfm?pub_id=957454
2. Avellaneda and Lee - Statistical arbitrage in the U.S. equities market - https://econpapers.repec.org/RePEc:taf:quantf:v:10:y:2010:i:7:p:761-782
3. TradingView - Paper Trading main functionality - https://www.tradingview.com/support/solutions/43000516466-paper-trading-main-functionality/
4. TradingView Pine docs - other timeframes and data - https://www.tradingview.com/pine-script-docs/concepts/other-timeframes-and-data/